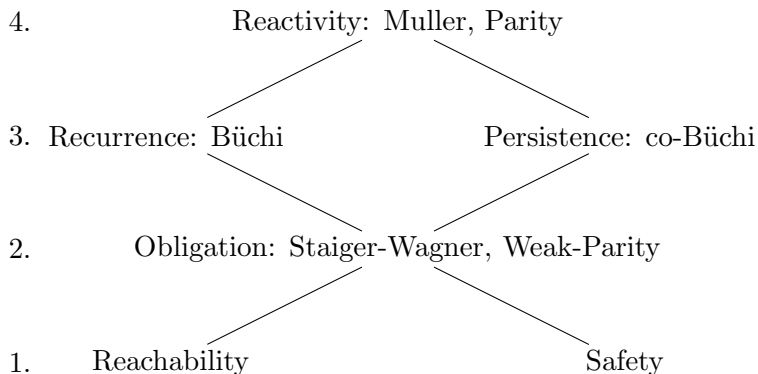


Obligation and Reactivity Games, Tree Automata

Hierarchy



Obligation Games

We consider games where the winning condition for Player 0 (on the play) is

- ▶ a Boolean combination of reachability conditions
- ▶ equivalently: a condition on the set Occ

Standard form: Staiger-Wagner winning condition $\mathcal{F} = \{F_1, \dots, F_k\}$

Player 0 wins play ρ iff $\text{Occ}(\rho) \in \mathcal{F}$.

We call these games **obligation games** (or **Staiger-Wagner games**).

Example

$$S = \{s_1, s_2, s_3\} \quad \mathcal{F} = \{\{s_1, s_2, s_3\}\}$$



No winning strategy is positional.

There is a finite-state winning strategy.

Weak Parity Games

Method for solving Staiger-Wagner games:

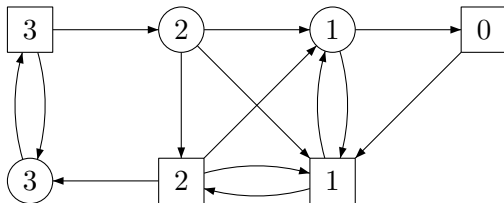
1. Solve weak parity games.
2. Reduce Staiger-Wagner games to weak parity games.

A **weak parity game** is a pair (G, p) , where

- ▶ $G = (S, S_0, E)$ is a game graph and
- ▶ $p : S \rightarrow \{0, \dots, k\}$ is a priority function mapping every state in S to a number in $\{0, \dots, k\}$.

A play ρ is winning for Player 0 iff the minimum priority occurring in ρ is even: $\min_{s \in \text{Occ}(\rho)} p(s)$ is even

Example



Weak Parity Games

Theorem

For a weak parity game one can compute the winning regions W_0 , W_1 and also construct corresponding positional winning strategies.

Proof.

Let $G = (S, S_0, E)$ be a game graph, $p : S \rightarrow \{0, \dots, k\}$ a priority function. Let $P_i = \{s \in S \mid p(s) = i\}$.

First steps if $P_0 \neq \emptyset$: We first compute $A_0 = \text{Attr}_0(P_0)$, clearly from here Player 0 can win.

In the rest game, we compute $A_1 = \text{Attr}_1(P_1 \setminus A_0)$ from here Player 1 can win.

General Construction

Aim: Compute $A_0, A_1, \dots, A_k$

Let G_i be the game graph restricted to $S \setminus (A_0 \cup \dots \cup A_{i-1})$.

$\text{Attr}_0^{G_i}(M)$ is the 0-attractor of M in the subgraph induced by G_i

$$A_0 := \text{Attr}_0(P_0)$$

$$A_1 := \text{Attr}_1^{G_1}(P_1 \setminus A_0)$$

for $i > 1$:

$$A_i := \begin{cases} \text{Attr}_0^{G_i}(P_i \setminus (A_0 \cup \dots \cup A_{i-1})) & \text{if } i \text{ is even} \\ \text{Attr}_1^{G_i}(P_i \setminus (A_0 \cup \dots \cup A_{i-1})) & \text{if } i \text{ is odd} \end{cases}$$

Correctness

Correctness Claim:

$$W_0 = \bigcup_{i \text{ even}} A_i \text{ and } W_1 = \bigcup_{i \text{ odd}} A_i$$

and the union of the corresponding attractor strategies are positional winning strategies for the two players on their respective winning regions.

Prove by induction on $i = 0, \dots, k$ the following:

$$\bigcup_{j=0\dots i, j \text{ even}} A_j \subseteq W_0 \text{ and } \bigcup_{j=1\dots i, j \text{ odd}} A_j \subseteq W_1$$

Correctness (cont.)

Base:

- ▶ **i=0:** $A_0 = \text{Attr}_0(P_0) \subseteq W_0$
- ▶ **i=1:** $A_1 = \text{Attr}_1^{G_1}(P_1 \setminus A_0) \subseteq W_1$

Induction step:

- ▶ **i even:** Consider play ρ starting A_i that complies to attractor strategy.
 - ▶ **Case 1:** ρ eventually leaves A_i to some A_j (from a Player-1 state), which $j < i$ and even, then Player 0 wins by induction hypothesis.
 - ▶ **Case 2:** ρ visits P_i , then we need to show that ρ visits only states with $p(s) \geq i$. Consider a state $s \in A_i$ that visits P_i , then
 - ▶ if $s \in S_0$, then not all edges lead to states with lower priority, otherwise $s \in A_j$ for some $j < i$. Contradiction.

Correctness (cont.)

- ▶ Case 2 (cont.):

- ▶ if $s \in S_1$, then all edges lead to states with priority $\geq i$. Any edge to a lower priority must lead to A_j with even j (Case 1). If there were edges to states s' with priority $j < i$ and j odd, then s' would already be in A_j . Contradiction.

- ▶ i odd: switch players

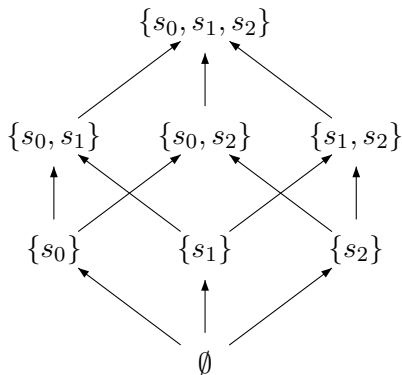
Obligation/Staiger-Wagner to Weak-Parity Games

- ▶ How to translate a Staiger-Wagner **automaton** to Weak-Parity automaton?
- ▶ Idea: record visited states during a run
- ▶ Record set: $R \subseteq S$
- ▶ Question: How to give priorities?

Record Sets and Priorities

Assume a game with states $\{s_0, s_1, s_2\}$.

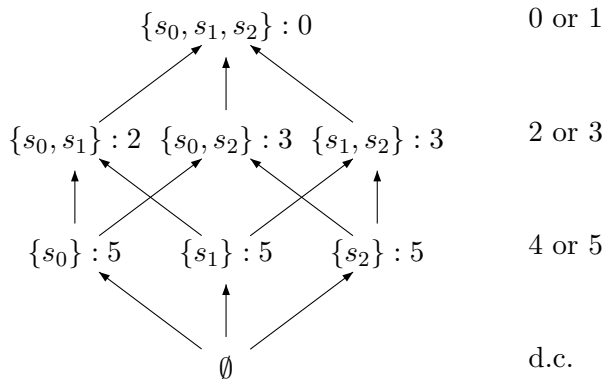
Consider possible record sets:



Assume the following play $s_1, s_0, s_1, s_0, s_2, \dots$ and the winning condition $F = \{\{s_0, s_1\}, \{s_0, s_1, s_2\}\}$. How to assign priorities?

Record Sets and Priorities

$F = \{\{s_0, s_1\}, \{s_0, s_1, s_2\}\}$. How would you assign priorities?



Idea of Game Reduction

We want to solve Staiger-Wagner games. We use a reduction to weak parity games (and the positional winning strategies of weak parity games).

Reduction will transform a game (G, ϕ) into a game (G', ϕ') such that usually

- ▶ G' is (usually) larger than G
- ▶ ϕ' is simpler than ϕ (so the solution of (G', ϕ') is simpler than that of (G, ϕ))
- ▶ from a solution of (G', ϕ') we can construct a solution of (G, ϕ) .

Concrete application: Transform Staiger-Wagner game into a weak parity game over a larger graph (from S proceed to $S \times 2^S$)

Game Reduction

Let $G = (S, S_0, E)$ and $G' = (S', S'_0, E')$ be game graphs with winning conditions ϕ and ϕ' , respectively.

(G, ϕ) is **reducible** to (G', ϕ') if:

1. $S' = S \times M$ for a finite set M and $S'_0 = S_0 \times M$
2. Each play $\rho = s_0 s_1 \dots$ over G is translated into a play $\rho' = s'_0 s'_1 \dots$ over G' by
 - ▶ a function $g : S \rightarrow S \times M$ (marks the beginning of ρ').
 - ▶ for all states $(s, m) \in S \times M$ in G' and all states $s' \in S$ in G , if there exists an edge $(s, s') \in E$, then there is a unique m' with $((s, m), (s', m')) \in E'$
 - ▶ for each edge $((s, m), (s', m')) \in E'$ in G' , there is an edge $(s, s') \in E$ in G
3. For all plays ρ and ρ' according to 2.: $\rho \in \phi$ iff $\rho' \in \phi'$

Application of Game Reduction

Theorem

Suppose (G, ϕ) is reducible to (G', ϕ') with extension set M , initial function g , and G and G' defined as before. Then, if Player 0 wins in (G', ϕ') from $g(s)$ with a memoryless winning strategy, then Player 0 wins in (G, ϕ) from s with a finite-state strategy.

Idea: Given a memoryless winning strategy $f : S'_0 \rightarrow S'$ from $g(s)$ for Player 0 in (G', ϕ') , we can construct a strategy automaton $A = (M, m_0, \delta, \lambda)$ for Player 0 in (G, ϕ) .

Obligation/Staiger-Wagner Games

Theorem

Given a Staiger-Wagner game (G, ϕ) , one can compute the winning regions of Player 0 and 1 and corresponding finite state strategies.

Proof.

We can apply game reduction with (G', ϕ') as follows:

$$G' := (S', S'_0, E')$$

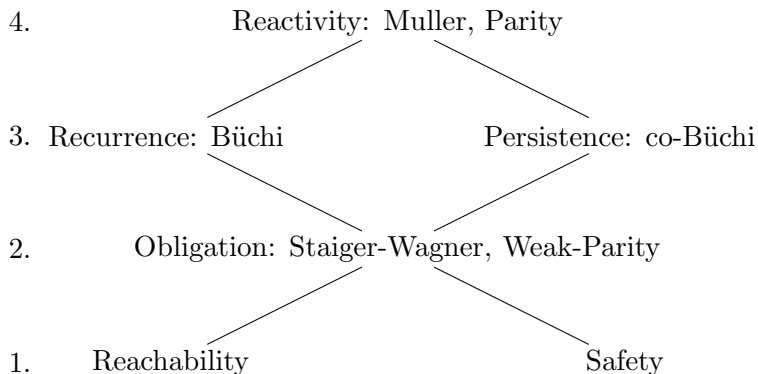
$$S' := S \times 2^S$$

$$((s, R), (s', R')) \in E' \quad \text{iff} \quad (s, s') \in E, R' = R \cup \{s'\}$$

$$g(s) = (s, \{s\})$$

$$p((s, R)) := 2 \cdot |S| - \begin{cases} 2 \cdot |R| & \text{if } R \in \phi \\ 2 \cdot |R| - 1 & \text{if } R \notin \phi \end{cases}$$

Hierarchy



Parity Games

A **Parity game** is a pair (G, p) , where

- ▶ $G = (S, S_0, E)$ is a game graph and
- ▶ $p : S \rightarrow \{0, \dots, k\}$ is a priority function mapping every state in S to a number in $\{0, \dots, k\}$.

A play ρ is winning for Player 0 iff the minimum priority visited infinitely often in ρ is even: $\min_{s \in \text{Inf}(\rho)} p(s)$ is even.

Parity Games

Theorem

1. *Parity games are determined (i.e., each state belongs to W_0 or W_1), and the winner has a positional winning strategy.*
2. *Over finite graphs, the winning regions and winning strategies of the two players can be effectively computed.*

Overview

We will show two proofs:

- ▶ One for general (even infinite) game graph
- ▶ One constructive for finite game graphs to establish effectiveness.

Proof 1

Given $G = (S, S_0, E)$ with priority function $p : S \rightarrow \{0, \dots, d\}$ and let $P_i = \{s \in S \mid p(s) = i\}$. We proceed by induction on the number of priorities.

► **Base case:** we either have an even or an odd priority

Proof 1

Given $G = (S, S_0, E)$ with priority function $p : S \rightarrow \{0, \dots, d\}$ and let $P_i = \{s \in S \mid p(s) = i\}$. We proceed by induction on the number of priorities

- ▶ **Base case:** we either have an even or an odd priority
- ▶ **Induction step:** we assume that the minimum priority k is even (otherwise switch the roles of players 0 and 1 below).

Let Π_1 be the set of vertices from which player 1 has a positional winning strategy.

Show that from each vertex in $S \setminus \Pi_1$, player 0 has a positional winning strategy.

Proof 1: Induction step

Consider the subgame with vertex set $S \setminus \Pi_1$. Then, $S \setminus \Pi_1$ defines a subgame. Why?

- ▶ **Case 1:** $S \setminus \Pi_1$ does not contain the minimal priority k .
Induction hypothesis applies.
- ▶ **Case 2:** $S \setminus \Pi_1$ contains vertices of minimal (even) priority.
Then, $S \setminus (\Pi_1 \cup \text{Attr}_0(P_k \setminus \Pi_1))$ defines a subgame

Proof 1: Induction step

Player 0 can guarantee that starting from a vertex in $S \setminus \Pi_1$ the play remains there.

Either the play stays in $S \setminus (\Pi_1 \cup \text{Attr}_0(P_k \setminus \Pi_1))$ from some point on, or it visits $\text{Attr}_0(P_k \setminus \Pi_1)$ infinitely often.

In the first case player 0 wins by induction hypothesis with a positional strategy, in the second case by infinitely many visits to the lowest (even) priority, also with a positional strategy.

Altogether: Player 0 wins from each vertex in $S \setminus \Pi_1$ with a positional strategy.

Proof 2

Given $G = (S, S_0, E)$ with S finite and priority function $p : S \rightarrow \{0, \dots, d\}$. We proceed by induction on the number of states denoted by n .

- ▶ **Base case:** we either have one Player-0 or Player-1 state with a selfloop (Note that every state in a game has at least one outgoing edge). Then the priority of the state determines if $S = W_0$ or $S = W_1$.
- ▶ **Induction step:** Let $P_i = \{s \mid p(s) = i\}$ be the set of states with priority i . Assume $P_0 \neq \emptyset$, otherwise assume $P_1 \neq \emptyset$ and switch the roles of Players 0 and 1 below. Finally, if $P_0 = P_1 = \emptyset$ decrease every priority by 2.

Proof (induction step cont.)

Choose $s \in P_0$ and let $X = \text{Attr}_0(\{s\})$. Note that $S \setminus X$ is a subgame with $< n$ states.

The induction hypothesis gives a partition of $S \setminus X$ into winning regions U_0 and U_1 for Player 0 and 1, respectively, and corresponding positional winning strategies.

- **Case 1: Player 0 can guarantee a transition from s to $U_0 \cup X$,**
i.e., if $s \in S_0$, then there exists $s' \in U_0 \cup X$ such that $(s, s') \in E$
or if $s \in S_1$, then for all $(s, s') \in E$, $s' \in U_0 \cup X$ holds.

Claim:

- (i) $U_0 \cup X \subseteq W_0$
- (ii) $U_1 \subseteq W_1$.

Proof (Case 1 cont.)

The positional strategy for Player 0 on $U_0 \cup X$ is:

1. On U_0 play according to the positional strategy given by the induction hypothesis
2. On $X (= \text{Attr}_0(\{s\}))$ play according to the attractor strategy.
Then eventually reach s
3. From s “move back” to $U_0 \cup X$ (by the assumption of Case 1).

For Player 1 use the positional strategy on U_1 given by the induction hypothesis.

Proof of claim: (ii) is clear, since starting in U_1 Player 1 can guarantee that the play remains in U_1 . For (i), the play remains in $U_0 \cup X$ if the strategy for state s is followed. If the play eventually remains in U_0 , then Player 0 wins by induction hypothesis, otherwise the play passes through s infinitely often, which is winning as well.

Proof (Case 2)

- **Case 2:** Player 1 can guarantee a transition to U_1 from s , i.e., if $s \in S_0$, then all edges $(s, s') \in E$ lead to U_1 ($s' \in U_1$), and if $s \in S_1$, then there exists $s' \in U_1$ such that $(s, s') \in E$.

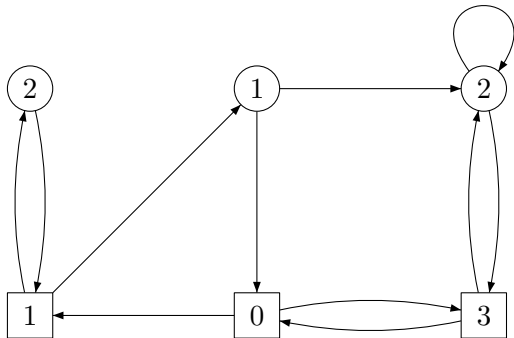
Let $Y = \text{Attr}_1(U_1)$, then $s \in Y$ and $S \setminus Y$ is a subgame with $< n$ states. The induction hypothesis gives winning region V_0 and V_1 and corresponding positional winning strategies.

Claim:

- (i) $V_0 \subseteq W_0$
- (ii) $V_1 \cup Y \subseteq W_1$.

Proof of claim: (i) is clear, since Player 0 can guarantee to stay within V_0 . For (ii), for all states in Y , Player 1 can guarantee to move to U_1 and stay there. From $t \in V_1$ Player 0 can either move to Y or stay in V_1 . Both choices are winning for Player 1.

Example



Recap

Winning conditions are defined over Occ and Inf.

Occ(ρ)	Inf(ρ)
Reachability/Guarantee game	Büchi game
Safety game	co-Büchi game
Weak-parity game	Parity game
Obligation/Staiger-Wagner game	Muller game

Recap

How did we solve those games?

Game	Solution
Reachability games	Attractor + Attractor Strategy
Safety games	like Reachability games
Büchi games	Recurrence set + Extended Attractor Strategy
co-Büchi games	like Büchi games
Weak-parity games	Alternation between Attr_0 and Attr_1
Obligation games	Reduction to Weak-parity games + record sets
Parity games	Recursive algorithm

Games and Tree Automata

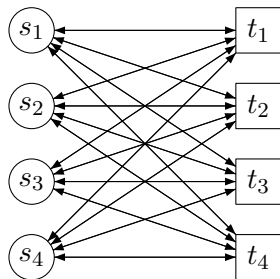
Muller Games

Given a game graph $G = (S, S_0, E)$ and a Muller condition $\mathcal{F} \subseteq \mathcal{P}(S)$, then a play ρ is winning for Player 0 if $\text{Inf}(\rho) \in \mathcal{F}$.

Recall, in Staiger-Wagner games, we had $\text{Occ}(\rho) \in \mathcal{F}$.

Example

Player 0 wins iff the number of states in $S_0 = \{s_1, s_2, s_3, s_4\}$ visited infinitely often is equal to the lowest index of the states in $S_1 = \{t_1, t_2, t_3, t_4\}$ visited infinitely often.



Winning condition in Muller form: $F \in \mathcal{F}$ iff $\min_i (t_i \in F) = |F \cap S_0|$.

Record the Past

For simplicity, we record only the s -states.

Visited letter	Record set
s_1	s_1
s_3	$s_1 s_3$
s_3	$s_1 s_3$
s_4	$s_1 s_3 s_4$
s_2	$s_1 s_2 s_3 s_4$
s_4	$s_1 s_2 s_3 s_4$
s_3	_" _
s_4	_" _
s_4	_" _

Last Appearance Record

Visited letter	Record set	LAR
s_1	s_1	$s_1 s_2 s_3 s_4 (1)$
s_3	$s_1 s_3$	$s_3 s_1 s_2 s_4 (3)$
s_3	$s_1 s_3$	$s_3 s_1 s_2 s_4 (1)$
s_4	$s_1 s_3 s_4$	$s_4 s_3 s_1 s_2 (4)$
s_2	$s_1 s_2 s_3 s_4$	$s_2 s_4 s_3 s_1 (4)$
s_4	$s_1 s_2 s_3 s_4$	$s_4 s_2 s_3 s_1 (2)$
s_3	_" -	..
s_4	_" -	..
s_4	_" -	..

Example

Assume the states s_3 and s_4 are repeated infinitely often but not s_1, s_2 . Then:

- ▶ the states s_1 and s_2 eventually arrive at the last two positions and are not touched any more, so finally the hit appears at most on positions 1 and 2
- ▶ position 2 is hit again and again; if only position 1 is hit from some point onwards, only the same letter would be chosen from there onwards (and not two states s_3 and s_4 as assumed)

Example

LAR-strategy for Player 0:

During play update and use the LAR as follows:

- ▶ shift the letter of the current state to the front
- ▶ record the position from where the current letter was taken
- ▶ move to the state whose index is the current hit position

This is a finite-state winning strategy with $n! \cdot n$ memory states if n letter states and n number states occur in the game graph.

From Muller to Parity Games

Theorem

For a game (G, ϕ) with $G = (S, S_0, E)$ and Muller winning condition ϕ (using the set $\mathcal{F} \subseteq 2^S$), there is a game (G', ϕ') with $G' = (S', S'_0, E')$ and parity winning condition ϕ' such that $(G, \phi) \leq (G', \phi')$

Proof.

Assume $S = \{1, \dots, n\}$. Define $S' := \text{LAR}(S)$

$\text{LAR}(S)$ is the set of pairs $((i_1, \dots, i_n), h)$ consisting of a permutation of $1, \dots, n$ and a number $h \in \{1, \dots, n\}$.

Construction

Initialisation: For $i \in S$ set

$$g(i) = ((i, i + 1, \dots, n, 1, \dots, i - 1), 1)$$

Definition of E' : Introduce an edge from $((i_1 \dots i_n), h)$ to $((i_m i_1 \dots i_{m-1} i_{m+1} \dots i_n), m)$ if $(i_1, i_m) \in E$

Construction

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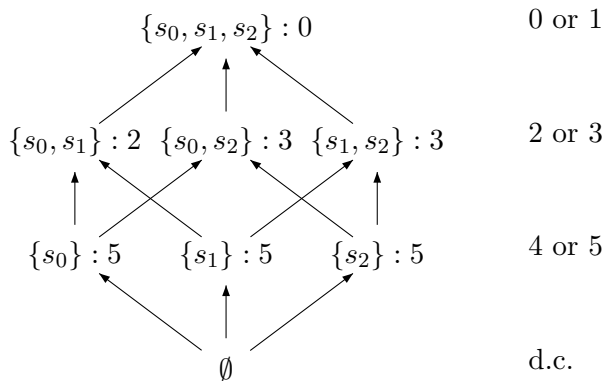
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How should we assign the priorities?

Record Sets and Priorities

Recall, priorities in the reduction of Staiger-Wagner to Weak-Parity.

$$F = \{\{s_0, s_1\}, \{s_0, s_1, s_2\}\}.$$



Construction(2)

Now, we are only interested in states visited infinitely often. The **hit value** tells us how many states are visited infinitely often.

E.g., if s_0 and s_1 are visited infinitely often, we see from some point on only the LARs: $(s_0s_1 \dots, 1), (s_0s_1 \dots, 2), (s_1s_0 \dots, 1), (s_1s_0 \dots, 2)$. If $\mathcal{F} = \{\{s_0, s_1\}\}$, then we want plays that visit only $(s_0s_1 \dots, 1)$ or $(s_1s_0 \dots, 1)$ from some point on to be losing. So, the priorities assigned to $(s_0s_1 \dots, 2)$ or $(s_1s_0 \dots, 2)$ need to override the priorities of $(s_0s_1 \dots, 1)$ or $(s_1s_0 \dots, 1)$.

Construction(2)

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Priorities $p : LAR(S) \rightarrow \{1, \dots, 2n\}$

$$p((i_1 \dots i_n, h)) = 2n - \begin{cases} 2h - 1 & \text{if } \{i_1 \dots i_h\} \notin \mathcal{F} \\ 2h & \text{if } \{i_1 \dots i_h\} \in \mathcal{F} \end{cases}$$

Proof of Correctness

Lemma

Given a play ρ in (G, ϕ) and its counterpart ρ' in (G', ϕ') , then $\text{Inf}(\rho) = F$ with $|F| = m$ iff

- 1. in ρ' the hit value is $> m$ only finitely often*
- 2. in ρ' the hit-segment is equal to F infinitely often*

Proof (forward).

Let $\text{Inf}(\rho) = F$ and $|F| = m$. Choose k and $k' > k$ s.t. for all $j > k$ $\rho(j) \in F$ and $\{\rho(k), \dots, \rho(k' - 1)\} = F$.

By construction of ρ' , the F -states $F = \{i_1, \dots, i_m\}$ are at the beginning of $\rho'(k')$ and for every $k'' > k'$ the hit is always $\leq m$ (1).

Proof of Correctness

Proof (forward cont.)

For the hit equal to m the hit-segment must be the set F . So, for (2) it suffices to show that the hit is infinitely often equal to m . Assume the hit is only finitely often equal to m , then eventually the LAR-entries $i_m, i_{m+1}, \dots, i_n$ are not changed anymore (and so, these states are not visited anymore). Then, $|\text{Inf}(\rho)| < m$, which contradicts $\text{Inf}(\rho) = F$ with $|F| = m$.

Proof (backwards).

Assume (1) and (2) holds. It follows from (1), that the LAR-entries $i_{m+1}, \dots, i_n$ in ρ' are fixed from some point j_0 onwards. So, the states $i_{m+1}, \dots, i_n$ are not visited anymore after j_0 . From, (2) it follows that $i_{m+1}, \dots, i_n$ are not in F (i.e., $\text{Inf}(\rho) \subseteq F$).

Proof of Correctness

Proof (backwards cont.)

For $F \subseteq \text{Inf}(\rho)$, assume $q \in F$ but $q \notin \text{Inf}(\rho)$.

Since $q \in F$ and $\text{hit-segment} = F$ infinitely often (2), we know that $q \in \text{hit-segment}$ infinitely often. Furthermore, since $|\text{hit-segment}| \leq m$ from some point on (1), it follows that from some point on the index i of q in the hit segment is $\leq m$. From $q \notin \text{Inf}(\rho)$ it follows that from some point onwards q can only stay in the same position in the LAR or go to the right and its final position i is $> m$. Contradiction.

Summary

We can solve Muller games by reduction to parity games using the **Last Appearance Record** construction.

Summary

We can solve Muller games by reduction to parity games using the **Last Appearance Record** construction.

Finally, **Rabin** games can be viewed as Muller games.

Back to Tree Automata

Muller tree automaton

Recall, a Muller tree automaton over Σ is $A = (S, s_0, T, \mathcal{F})$, where

- ▶ S is a finite set of states,
- ▶ $s_0 \in S$ is an initial state,
- ▶ $T : S \times \Sigma \rightarrow 2^{S \times S}$ is a transition function
- ▶ $\mathcal{F} \subseteq 2^S$ is the set of accepting sets.

Given an input tree t , a run π of A over t is **accepting** iff for every path σ in t :

$$\text{Inf}(\pi|_{\sigma}) \in \mathcal{F}$$

Parity tree automaton

A Parity tree automaton over Σ is $A = (S, s_0, T, p)$, where

- ▶ S is a finite set of states,
- ▶ $s_0 \in S$ is an initial state,
- ▶ $T : S \times \Sigma \rightarrow 2^{S \times S}$ is a transition function
- ▶ $p : S \rightarrow \{0, \dots, k\}$ is a priority function.

Given an input tree t , a run π of A over t is **accepting** iff for every path σ in t :

$$\min_{s \in \text{Inf}(\pi|_{\sigma})} p(s) \text{ is even}$$

Example

A parity tree automaton over $\Sigma = \{a, b\}$ that recognizes all binary trees

$$\mathcal{T} = \{t \in \mathcal{T}^\omega(\Sigma) \mid \text{each path through } t \text{ has only finitely many } b\}$$

- ▶ $S = \{q_a, q_b\}$
- ▶ $I = \{q_a, q_b\}$
- ▶ $T(q_a, a) = \{(q_a, q_a)\}, T(q_b, a) = \{(q_a, q_a)\}$
 $T(q_a, b) = \{(q_b, q_b)\}, T(q_b, b) = \{(q_b, q_b)\}$
- ▶ $p(q_a) = 2, p(q_b) = 1$

Parity Automata $\leftrightarrow$ Muller Automata

Theorem

1. *For any parity tree automaton one can construct an equivalent Muller tree automaton.*
 2. *For any Muller tree automaton one can construct an equivalent parity tree automaton.*
-
1. Given a parity tree automaton $A = (S, s_0, T, p)$ keep states and transitions and define $\mathcal{F}$ as follows:

$$\mathcal{F} = \{F \in 2^S \mid \min_{s \in F} p(s) \text{ is even}\}$$

Parity $\leftrightarrow$ Muller

2. Copy the simulation of Muller games by parity games. Given a Muller tree automaton with state set S use for the parity tree automaton the state set $\text{LAR}(S)$ and define the transition according to the LAR update rule.

Allow transition

$$((s_1 \dots s_n, i), a, (s'_1 \dots s'_n, j), (s''_1 \dots s''_n, k))$$

for transition (s_1, a, s'_1, s''_1) of the Muller automaton, where

- ▶ $(s'_1 \dots s'_n, j)$ is the LAR update for a visit to s'_1 and
- ▶ $(s''_1 \dots s''_n, k)$ is the LAR update for a visit to s''_1 .

Define priorities as in the simulation of Muller games by parity games.

Tree Automata and Games

With any parity tree automaton $A = (S, s_0, T, p)$ over Σ and any input tree $t \in \mathcal{T}^\omega(\Sigma)$, we can associate a parity game between

- ▶ Player **Automaton** and
- ▶ Player **Pathfinder**

that proceeds as follows:

- ▶ First, Automaton picks a transition in T (from s_0) which matches the labels of the root of t
- ▶ Then Pathfinder decides on a direction (left or right) to proceed to a son of the root
- ▶ Then Automaton chooses again a transition for this node (and compatible with the first transition)
- ▶ Then Pathfinder reacts again by branching left or right...

Tree Automata and Games

Such a play give a sequence of transitions (and hence a sequence of states in S) built up along a path chosen by Pathfinder.

Automaton wins the play iff the sequence of states satisfies the parity condition.

Given a parity tree automaton $A = (S, s_0, T, p)$ over Σ and an input tree t , the *game graph* $G_{A,t} = (S_0 \cup S_1, S_0, E)$ is defined by

- ▶ $S_0 = \{(w, t(w), s) \mid w \in \{0, 1\}^*, t(w) \in \Sigma, s \in S\},$
- ▶ $S_1 = \{(w, t(w), \tau) \mid w \in \{0, 1\}^*, t(w) \in \Sigma, \tau \in T\},$

and the edges relation E is such that successive game positions are compatible with the transitions in A on t .

The priority of a triple $u = (w, t(w), s)$ or $(w, t(w), (s, t(w), s', s''))$ is the priority $p(s)$. (Standard initial position: $(\epsilon, t(\epsilon), s_0)$)

Tree Automata and Games

Lemma

The tree automaton A accepts an input tree t iff in the parity game over $G_{A,t}$ there is a winning strategy for player Automaton from the initial position $(\epsilon, t(\epsilon), s_0)$.

Proof.

A successful run of A on t yields a winning strategy for Automaton in the parity game over $G_{A,t}$: Along each path the suitable choice of transitions is fixed by the run.

Conversely, a winning strategy for Automaton over $G_{A,t}$ clearly provides a method to build up a successful run of A on t . Just apply this winning strategy along arbitrary paths.

Summary: Tree Automaton

- ▶ Tree Automata can be viewed as games between **Automaton** and **Pathfinder**
- ▶ Parity and Muller tree automata can be reduced to each other
- ▶ (Same holds for Rabin/Streett, Parity, and Muller tree automata)
- ▶ We showed closure properties of Muller tree automaton (union, intersection, projection)
- ▶ Missing: complementation

Complementation of Parity Tree Automaton

We will show basic idea.

- ▶ To complement a given automaton A means to construct an automaton B s.t.

$$t \notin A \leftrightarrow t \in B$$

- ▶ Due to the run lemma, complementation means to conclude from the **non-existence** of a winning strategy of Player Automaton in $G_{A,t}$ that there exists a winning strategy of Automaton in $G_{B,t}$.

Proof has two steps:

1. use determinacy of parity games to show that if Automaton has no winning strategy over $G_{A,t}$, then Pathfinder has a winning strategy over $G_{A,t}$ (from $(\epsilon, t(\epsilon), s_0)$)
2. Convert Pathfinder's strategy into an Automaton strategy.

Complementation of Parity Tree Automaton

Theorem

*For any parity tree automaton A over Σ , **one can build** a Muller tree automaton (and therefore a parity tree automaton) B over Σ that recognizes $\mathcal{T}^\omega(\Sigma) \setminus \mathcal{L}(A)$*

Proof.

From Step 1 (determinacy of parity games), we know there exists a (memoryless) winning strategy $f : S_1 \rightarrow \{0, 1\}$ for Player Pathfinder.

$$f : \{0, 1\}^* \times \Sigma \times T \rightarrow \{0, 1\}$$

decompose f into a family of strategies parameterized by $w \in \{0, 1\}^*$

$$f_w : \Sigma \times T \rightarrow \{0, 1\}$$

Complementation of Parity Tree Automaton

Let I be the set of all possible **local instructions** $i : \Sigma \times T \rightarrow \{0, 1\}$.

Then, f can be represented as I -labeled binary tree s with $s(w) = f_w$.

Let $s \cdot t$ be the corresponding $(I \times \Sigma)$ -labeled tree

$$(s \cdot t)(w) = (s(w), t(w)) \text{ for } w \in \{0, 1\}^*.$$

Since f exists, we know there is an I -labeled tree s s.t. for all sequences $\tau_0\tau_1 \dots$ of transitions chosen by Automaton and for all paths (in path for the unique) $\pi \in \{0, 1\}^*$, the generated state sequence violates the parity condition.

Intuitively, f tells the “new” automaton for every tree $t \notin \mathcal{L}(A)$ which path to track for a given transition sequences in order to reject/accept the tree t .

Complementation of Parity Tree Automaton

So, we know:

1. There exists an I -labeled tree s such that $s \cdot t$ satisfies
2. for all $\pi \in \{0, 1\}^\omega$
3. for all $\tau_0\tau_1 \dots \in T^\omega$
4. if the sequence $s|_\pi$ of local instructions applied to the sequence of tree labels $t|_\pi$ and the sequence $\tau_0\tau_1 \dots$ produces the path π , then the state sequence determined by $\tau_0\tau_1 \dots$ violates the parity condition.

Complementation of Parity Tree Automaton

- ▶ Condition 4 is a property of ω -words over $I \times \Sigma \times T \times \{0, 1\}$, which can be checked by a Muller word automaton M_4 .
- ▶ Condition 3 is a property of ω -words over $I \times \Sigma \times \{0, 1\}$ checked by M_3 , which results from M_4 by universally quantifying T (negate, project, negate).
- ▶ Condition 2 is a property of $(I \times \Sigma)$ -labeled trees, which can be checked by a Muller tree automaton M_2 that simulates M_3 along each path.
- ▶ Condition 1, apply nondeterminism, a Muller tree automaton B can be built by guessing a tree s on the input tree t and running M_2 on $s \cdot t$.